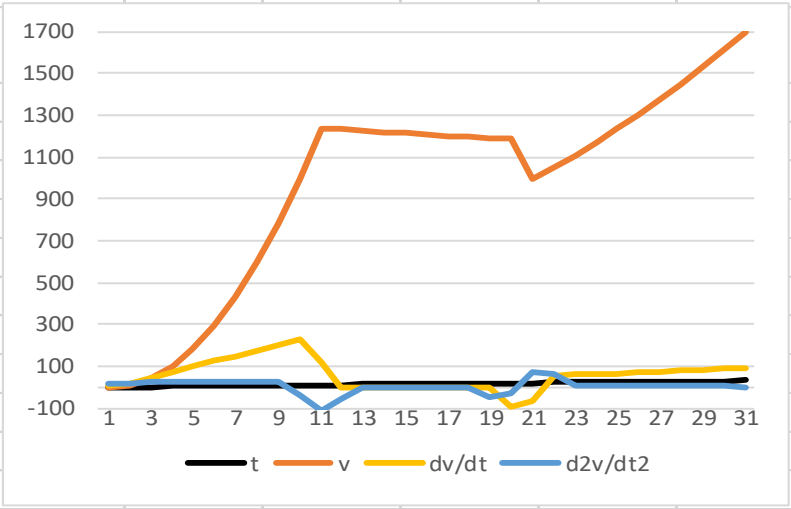
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1. (20pts) Employ the multiple-application Simpson's rule to evaluate the vertical distance traveled by a rocket if the vertical velocity is given by

$$\begin{aligned} v &= 13t^2 - 6t & 0 \le t \le 10 \\ v &= 1300 - 6t & 10 \le t \le 20) \\ v &= 50t + 2(t-20)^2 & 20 \le t \le 30 \end{aligned}$$

In addition, use numerical differentiation to develop graphs of the acceleration (dv/dt) and the jerk (d^2v/dt^2) versus time for t = 0 to 30. Note that the jerk is very important because it is highly correlated with injuries such as whiplash.

t	v	Mult. Simp 1/3 rule	dv/dt	d^2v/dt^2
0	0	29240	7	13
1	7		20	19,5
2	40		46	26
3	99		72	26
4	184		98	26
5	295		124	26
6	432		150	26
7	595		176	26
8	784		202	26
9	999		228	-42,25
10	1240		117,5	-117
11	1234		-6	-61,75



2. (20pts) Solve from t = 0 to 3 with h = 0.1 using Euler's method,

$$\frac{dy}{dt} = y \sin^3 t\,, \quad y(0) = 1$$

t	y	dy/dt
0	1	0
0,1	1	0,000995
0,2	1,0001	0,007842
0,3	1,000884	0,025831
0,4	1,003467	0,059259
0,5	1,009393	0,11123
0,6	1,020516	0,183713
0,7	1,038887	0,277758
0,8	1,066663	0,39376

Note: **your answer sheet should involve script, result and your comments.** Good Luck

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t	y	dy/dt				
0	1	0				
0,1	1	0,000995				
0,2		0				
0,3		0				
0,4		0				
0,5		0				
0,6		0				
0,7		0				

3) (20pts) Use Gauss Elimination to solve the system.

$$\begin{bmatrix}1 & 2 & 3\\ 3 & 4 & 5\\ 5 & 6 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}2\\ 4\\ 6\end{bmatrix}$$

4) (20pts) Solve the system Ax=b using Gauss Jordan Elimination

$$A=\begin{bmatrix}2 & 1 & -1\\ 1 & 3 & 4\\ -2 & 5 & 2\end{bmatrix},\quad b=\begin{bmatrix}0\\ 9\\ 2\end{bmatrix}$$

5) (20pts) Find out the Lagrange interpolating polynomial of the first, second and third order to evaluate f(0.5) on the basis of the data given.

xi	-1	0	1	2
f(xi)	1	1	1	-5

Note: **your answer sheet should involve script, result and your comments.** Good Luck